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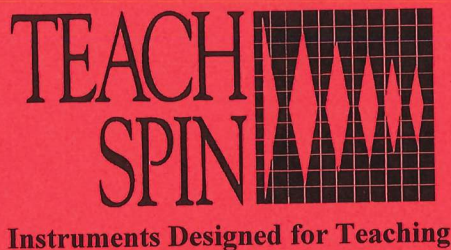
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FARADAY ROTATION

Guide to the Experiment

INSTRUCTOR'S MANUAL

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FARADAY ROTATION

Instructor's Guide to TeachSpin's FR1-A Apparatus

by

Jonathan F. Reichert, Ph.D.

I. INTRODUCTION

"Impelled by a belief in the unity of the forces in nature, Michael Faraday sought, and in 1845 provided, the first phenomenological evidence for the connection between light and magnetism when he discovered the effect that still bears his name. He found that plane-polarized light, propagating through matter parallel to a static magnetic field, underwent a systematic rotation of its plane of polarization."

Thus begins an interesting paper by David A. Van Baak on the topic of Faraday rotation in atomic gas.¹ Although carefully studied from the time it was discovered, this remarkable effect was not theoretically understood until the development of quantum mechanics. References in Van Baak's paper provide some insight into current efforts both experimentally and theoretically, even for nonlinear optics. Many textbooks also discuss this topic.²

Although the effect is typically small, Faraday rotation has found important practical applications. In modern fiber optic communications networks, it is used as the canonical optical isolator.³ This is an extremely important component of such networks since it prevents the multiple propagation of reflected signals. Faraday rotation has important applications in optical modulators as well as uses in analytic apparatus for mixtures of hydrocarbons.

We strongly urge the instructor to make use of these references. Copies of two American Journal of Physics articles have been included in this manual to facilitate student exploration of the literature on the Faraday effect. Although it is an important effect, it is frequently omitted in the standard courses in E&M and Optics in both the undergraduate and graduate physics and engineering curriculum.

TeachSpin originally thought of this apparatus in connection with our teaching Signal Processor/Lock-In Amplifier. It seemed to us that the Faraday effect was an ideal experiment to show both interesting physics and the power of lock-in detection for extracting weak signals from background noise. It was with this in mind that we designed FR1-A. However, in the direct DC mode, the unit can certainly be used without lock-in detection. It must be noted that this unit comes with only one monochromatic light source, a laser pointer with a wavelength of about 650 nm. Instructors who want the student to explore the interesting frequency dependence of the effect will have to supply their own polarized light sources at other wave lengths. The Polaroid film analyzer has a useful polarizing capability from 400 to 800 nm. Since the Faraday effect is a slowly varying function of wavelength, the intensity stability of the light source is more important than its frequency stability. The diode laser is an ideal source for these measurements.

II. APPARATUS

FR1-A consists of four basic components: the Light Source, the Solenoid (magnetic field source), the Analyzer Polaroid and the Optical Detector. Each one will be described separately.

A. The Light Source

A red laser pointer is used as the light source. It operates at a nominal wavelength of approximately 650 nm with a power output of about 3 mw. Although the output is approximately 60% polarized, the laser light is directed through a polarizing filter which increases its polarization to about 95%. Since this filter is not permanently bonded to the laser, it is important to adjust its rotation position to achieve maximum power output of the light source. This will occur when the polarization axis of the laser corresponds to that of the filter. This can easily be adjusted visually by inserting a white card at the entrance to the solenoid.

The laser requires a voltage regulated supply of 4 volts and 40 mA. ***The red banana plug should be connected to the positive terminal post of the supply. Reversing the power supply voltage will destroy the laser.*** Note: it is important to protect both leads to the laser from an electrostatic shock. Keep the connectors shorted to each other until they are connected to their supply. TeachSpin's audio amplifier unit AALS1-A has a voltage regulated (4V) supply output specifically designed for this laser diode light source.

The laser beam must be adjusted by the operator to traverse the solenoid along its central axis. This can be accomplished by manipulation of the four nylon thumb screws on the laser mount using a white card both at the entrance and exit of the solenoid. The sample glass rod, or other samples, should be removed from the solenoid during the alignment procedure. After alignment, the sample can be reinserted into the solenoid.

B. Solenoid

The solenoid has the following physical specifications:

Length: 150 mm	Wire Size: #18 double insulated
Turns/layer: 140	DC Resistance: 2.6 Ω
Number of Layers: 10	

The approximate calibration constant for the unit, measured at the center of the solenoid is:

$$B = (11.1 \text{ mT/A}) I$$

where I is in amperes and B is in millitesla.

The magnetic field does vary along the axis of the solenoid. Such variation may be significant for certain samples, particularly those which extend outside the coils. We recommend that the students measure the axial magnetic field profile using TeachSpin's Hall Effect Probe HE1-A, or other equivalent field measuring apparatus.

The maximum continuous current through the unit is 3 amperes. For times of the order of 30 seconds, however, 10 amperes can be used without damaging the solenoid or its supports. **Extreme care** should be taken to assure that the unit does not overheat when exceeding the 3 A specification.

Since Faraday rotation measurements can easily be performed in 10-15 seconds by a student familiar with the equipment, it is possible to use these larger solenoid currents and thus obtain larger rotations.

C. Analyzer Polaroid

The unit is equipped with a rotatable Polaroid film in a calibrated mount. The decal is marked in 5° increments. This limits the accuracy of an angular measurement to about 2° . This is a conservative estimate. Some students may claim 1° set-ability. You be the judge.

D. Detector

The detector is simply a photodiode connected in series to one of three resistors; 10K, 3K, 1K. The photodiode is a *current source* and is a linear photonic detector, as long as the voltage across it is less than about 0.3 volt. Saturation begins to occur when this bias voltage appears across the diode. This makes the detector nonlinear.

It is wise to work at bias voltages well below the 0.3 volt value. This can be accomplished by using the proper load resistor and monitoring the bias voltage. Since the laser diode light source is high intensity, it will be necessary to use the lower values of load resistors for some Faraday rotation experiments.

E. Sample

FR1-A comes with one glass rod sample. Its specifications are:

Diameter: 5 mm	Length: 100 mm
Material: SF - 59	

The sample can be removed from the solenoid by first removing the detector, analyzer and light source. This gives the experimenter unrestricted access to the solenoid's bore. Care should be taken not to damage the end of the glass sample. They are highly polished. Any push-rod should have a soft end covered by optical tissue or its equivalent.

III: EXPERIMENTS

A. Malus's Law

With the sample removed from the solenoid, students can measure the intensity of the transmitted light through two polarizers (one on the laser, the other the analyzer) as a function of the relative angle between them. They should also derive a simple theoretical calculation of this "law."

Be Careful!! This is a simple experiment, but one must not saturate the diode detector. Use the correct load resistor.

B. Measure the Verdet Constant, v , DC.⁴

The Verdet constant is defined by the relationship:

$$\theta = vB\ell$$

where B is the strength of the magnetic field in millitesla, ℓ is the length in centimeters and θ , in radians, is the angle of rotation of the axis of the polarization after passing through the medium in the magnetic field. The units of v are then: radians/millitesla centimeter. These are by no means universal units. The units given in the literature vary and include degrees/gauss centimeter.

First, you may wish to independently measure the magnetic field inside the solenoid rather than using the $B = 11.1I$ calibration we have supplied. The field inside is not constant along the entire length. It drops off to about half the maximum value at the ends. TeachSpin's HE1-A Hall Effect Probe is an ideal instrument to measure this axial field, but other Hall sensor probes will also do the job. Such a calibration will allow the student to make corrections to their determination of the Verdet constant.

Place the 10 cm glass rod inside the solenoid, supporting it on the foam rubber sleeves. Make sure it is located in the center. Using any white surface (a business card will do) align the laser beam so that the light enters and exits along the center axis of the solenoid. It is important that this be done carefully so that the entire laser beam passes through the glass rod.

Now there are several ways the Faraday rotation can be measured. Both of the methods given below should be tried by the students so that they can explain which one is more accurate and why.

1. Method 1 – Extinction

- Set the analyzer polarizer at 90° with respect to the entrance polarizer. This gives maximum extinction of the light to the detector.
- Connect a current regulated supply to the solenoid.
- Measure the angle of rotation of the analyzer necessary to return to maximum extinction when the solenoid's magnetic field is turned on.
- Find the rotation angle for a variety of magnetic fields.

Describe what happens when you reverse the direction of the magnetic field?
Does the magnetic field affect the photodiode detector itself?

2. Method 2 – Returning to Fixed Intensity Value

- a. Set the analyzer at 45° with respect to the entrance polarizer.
- b. Repeat the procedure described in Method 1. This time, however, adjust the analyzer angle so that the detector output returns to the value found before the magnetic field is turned on.
- c. How does the value of the initial angle affect the results?

It should be straightforward to determine the Verdet constant from either of these measurements. Which method is best? Why?

C. Measure the Verdet Constant AC^5

A more accurate and more sensitive method of determining Verdet constants uses lock-in detection and ac magnetic fields. TeachSpin originally developed FR1-A to accompany its teaching Signal Processor/Lock-In Amplifiers (SPLIA1-A) since Faraday rotation was a good example of a “small” physical effect that can easily be measured with lock-in detection techniques. This was also independently realized by Jain etc. all who have written an excellent article on how this measurement is carried out. We have enclosed a copy of their American Journal of Physics article for your convenience.

Lock-In detection affords some additional possibilities for the student.

1. The relative angle at which to set the polarizer to obtain the maximum signal is easily determined.
2. The relative phase angle of the ac signal as a function of Polaroid angle can be examined.
3. The student develops an understanding of the process of obtaining an absolute measurement of v from these ac measurements.

D. Measurements of Verdet Constants in Liquids

The ac lock-in detection method allows one to detect extremely small Faraday rotations and thus small Verdet constants without using extremely large magnetic fields. Various liquids exhibit the Faraday effect and their constants can be measured with this apparatus. A glass cell with flat optical ends and a filling tube at one end will hold an appropriate liquid sample. With such a sample holder, however, the liquid will not all be in the same magnetic field. Part of the sample will even be outside the solenoid.

This presents an interesting challenge for the student's determination of the Verdet constant. It requires that the axial field profile be measured first, and a simple computer program or hand held calculator algorithm be developed to extract v from the measurements. This can easily be done since the Faraday effect is a linear function of the magnetic field.

IV. REFERENCES

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FARADAY ROTATION

Selected Readings

A simple experiment for determining Verdet constants using alternating current magnetic fields

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A simple experiment suitable for a senior undergraduate and graduate laboratory on the measurement of Faraday rotation using ac magnetic fields is described. The apparatus was used to measure the wavelength dependence of Verdet constants for several materials. A concentration-dependent study on ferric chloride water solution was also carried out to demonstrate that the diamagnetic and paramagnetic contributions to Faraday rotation have opposite signs.

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I. INTRODUCTION

Faraday rotation¹ is the rotation of the plane of polarization of light due to magnetic-field-induced circular birefringence in a material. In a nonabsorbing or weakly absorbing medium a linearly polarized monochromatic light beam passing through the material along the direction of the applied magnetic field experiences circular birefringence, which results in rotation of the plane of polarization of the incident light beam. The angle of rotation θ can be expressed as $\theta = VBL = \pi \Delta n L / \lambda$, where Δn is the magnitude of circular birefringence (a difference in refractive index of left and right circularly polarized light in the medium), L is the length of the medium traversed, λ is the wavelength of light in vacuum, and B is the magnetic field. The constant V is a material property and gives a quantitative measure of the Faraday rotation ability of the material and is called the Verdet constant. Faraday rotation is commonly used in fabrication of optical isolators² to prevent unwanted backreflections. Faraday rotation measurements can also be used to infer the susceptibility of materials and measure carrier densities³ in semiconductors if the effective mass of the carrier is known.

Many investigations have been carried out to measure the Verdet constants of materials using dc magnetic fields.⁴⁻⁷ Since Faraday rotation combines many fundamental elements of polarization optics with electromagnetism, dc magnetic-field-based Faraday rotation experiments have also been introduced to the undergraduate physics laboratory to examine the Faraday rotation and determine the Verdet constants of samples.^{5,6} However, the dc magnetic-field-based measurement requires an expensive and bulky electromagnet and a costly high current power supply to produce magnetic fields of the order of a kilogauss or higher due to the small values of the Verdet constants for paramagnetic and diamagnetic materials (of the order 10^{-3} to 10^{-2} min/G cm). Measurement of the Verdet constants with an ac magnetic field has been recently reported.⁷ In this paper, we describe a simple experimental setup suitable for a senior or graduate laboratory to investigate Faraday rotation and determine the Verdet constant using an ac magnetic field. The experiment uses a lock-in amplifier to measure the Faraday rotation angle and introduces students to the utility of the phase sensitive detection technique to extract a signal normally buried

in noise. The Verdet constants of several materials commonly available in the laboratory were determined at different wavelengths and compared with published data.^{8,9}

II. EXPERIMENT

The experimental setup is schematically shown in Fig. 1. The conventional large electromagnet is replaced with two 500-turn coils (wound with enamelled copper wire 1 mm in diameter). These coils were routinely used in our undergraduate labs for demonstrating concepts relating to inductances and resonant circuits. Each coil has an air core with a cross section of 4 cm $\times$ 4 cm and a length of 7 cm. Each of these coils has an inductance of 0.005 H and a resistance of 2.5 Ω allowing a maximum current of 2.5 A. The spacing between the coils is 3 cm. The power source to drive each of the coils generating the ac field was a car stereo amplifier capable of supplying 50-W per channel and was bought from a discount store for \$30.00. A 13.8-V 10-A power supply was used as the power source for the amplifier. Two 1- μ F capacitors (rated for 300 V) were connected in series with each of the coils to resonate the circuits at a convenient drive frequency which was chosen to be in the vicinity of 2 kHz. The resonant circuit helps cancel current limiting due to the inductive impedance of the coils and makes it possible to generate larger ac magnetic fields. It is important to bear in mind that due to the resonant excitation the voltage across

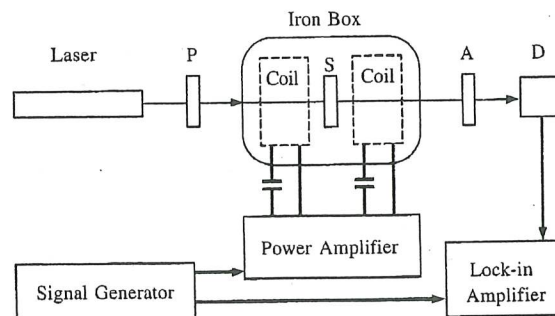


Fig. 1. Schematic diagram of the setup for measuring the Faraday rotation using ac magnetic field. P: polarizer; S: sample; A: analyzer; D: photodetector.

the capacitor and inductors can get as high as 200 V and care should be taken to make sure that these components are appropriately insulated or contained in a closed box to avoid a possible shock during operation. The sinusoidal output of a signal generator was fed into the two input channels of the audio amplifier and the frequency was scanned until resonance was obtained at 2.2 kHz. Typically, the ac magnetic field was adjusted to be in the range of 20–40 G rms. The magnitude of the magnetic field can be controlled by adjusting the magnitude of the input signal or the frequency of the signal generator so as to move slightly away from the resonance peak. The magnetic field was measured using a Hall probe ac gaussmeter. If a gaussmeter is not available, the magnetic field can still be determined by measuring the Faraday rotation of a material whose Verdet constant is accurately known. We shall further elaborate on this in Sec. IV.

In this experiment, a polarizer (inexpensive polarizers with extinction ratios of 1:1000 work fine) is used to obtain a linearly polarized light beam that passes through the sample placed between the two coils. The plane of polarization of the light beam passing through the sample under the influence of the ac magnetic field oscillates periodically. This oscillation of the plane of polarization is converted by another polarizer (as analyzer) that is set at 45° with respect to the input polarizer to an oscillation of light intensity which is detected by a photodetector and a lock-in amplifier (Stanford Research Systems SR510 single-phase analog lock-in amplifier with a frequency range of 0.5 Hz–100 kHz). The photodetector used here is a silicon-based low-power detector (Newport 818-SL) which has a broad spectral response of 400–1100 nm and a rise time about 2 μ s. The Faraday rotation angle is experimentally measured and the Verdet constant can be easily determined if the value of the magnetic field and sample thickness are known. Section III discusses some of the theoretical aspects underlying the experiment.

III. THEORY

The linearly polarized light incident on the sample is assumed to be propagating in the z direction and polarized along the x axis. The electric field of the light beam can be expressed in Jones matrix form as

$$E_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} A_0 \exp(-i\omega t + ikz), \quad (1)$$

where A_0 is the amplitude of the electric field of the beam. After passing through the sample, the polarization of the light is rotated by a small angle θ and can be represented by

$$E = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} A_0 \exp(-i\omega t + ikz). \quad (2)$$

The light then traverses through an analyzer which is set at an angle of ϕ with respect to the polarizer. The electric field of the light beam is now described as

$$E = \begin{pmatrix} \cos(\phi - \theta) \cos \phi \\ \cos(\phi - \theta) \sin \phi \end{pmatrix} A_0 \exp(-i\omega t + ikz) \quad (3)$$

and the light intensity measured by the detector is given by

$$I = \cos^2(\phi - \theta) A_0^2. \quad (4)$$

In order to achieve a maximum modulation of the light intensity (ΔI), the analyzer needs to be set at an optimal angle. By taking the first derivative of I with respect to θ , one gets

$$\frac{\partial I}{\partial \theta} = \sin 2(\phi - \theta) A_0^2. \quad (5)$$

Since the angle θ is very small ($\theta \ll 1^\circ$), one can clearly see that when $\phi = 45^\circ$ maximum ΔI can be obtained. With the analyzer set at 45°, the intensity measured by the detector as expressed by Eq. (4) can be simplified:

$$I = \frac{1}{2}(1 + 2\theta) A_0^2. \quad (6)$$

Since the ac magnetic field is sinusoidal $B = B_0 \sin(\Omega t)$ and θ is proportional to B , the rotation angle can be written in the form of $\theta = \theta_0 \sin(\Omega t)$. Then Eq. (6) can be rewritten as

$$I = \frac{1}{2}(1 + 2\theta_0 \sin(\Omega t)) A_0^2 = I_0 + \Delta I \sin(\Omega t). \quad (7)$$

By measuring the relative change of the light intensity $\Delta I/I_0$, the amplitude of the angle of rotation can be easily determined to be

$$\theta_0 = \frac{1}{2} \frac{2\theta_0 A_0^2}{A_0^2} = \frac{1}{2} \frac{\Delta I}{I_0}. \quad (8)$$

The Verdet constant of the material is then determined using the relation $\theta_0 = VB_0L$, where θ_0 , L , and B_0 are experimentally determined. It is important to point out that it is not difficult to measure a ratio $\Delta I/I_0$ of the order 10^{-5} using a phase sensitive technique. A rotation of the plane of polarization of the order of 10^{-5} rad is thus measurable by this technique due to the elimination of broadband noise by the lock-in amplifier. A rotation of 10^{-5} rad corresponds to a difference of the order 10^{-10} in the refractive index of left and right circularly polarized light in the medium.

IV. RESULTS

Experiments were performed on a number of materials which are relatively easily available in a teaching laboratory. The materials investigated included distilled water, ethyl alcohol, methyl alcohol, BK7 glass, and aqueous solutions of ferric chloride at different concentrations. For liquid samples, a glass cuvette with 1-cm path length was used. The thickness of the BK7 glass used in our experiments was 1.2 cm. Lasers were used as light sources for the wavelength-dependent experiments simply because they were available, but a collimated white light source followed by an appropriate wavelength selective interference filter works fine. The wavelengths used in these experiments were obtained from an argon-ion laser (458, 476, 488, 496, and 514 nm), a diode pumped solid state laser (532 nm), a He-Ne laser (633 nm), and two diode lasers (670 and 830 nm).

Previously published Verdet constants of water at different wavelengths⁸ were used as a standard to calibrate the magnetic field and check the results obtained with the gaussmeter. The rotation of the plane of polarization in distilled water as a function of the current passing through the coils was measured first and compared with previously published data to obtain an independent calibration of the magnetic field as a function of current. The wavelength-dependent results for distilled water are shown in Fig. 2(a) along with the published data from Ref. 8. The Verdet constants of ethyl alcohol, methyl alcohol, and BK7 glass were also measured and are shown in Fig. 2 (b), (c), and (d), respectively. The contribution from the glass cuvette has been subtracted for all liquid samples. The experimental error for the Verdet con-

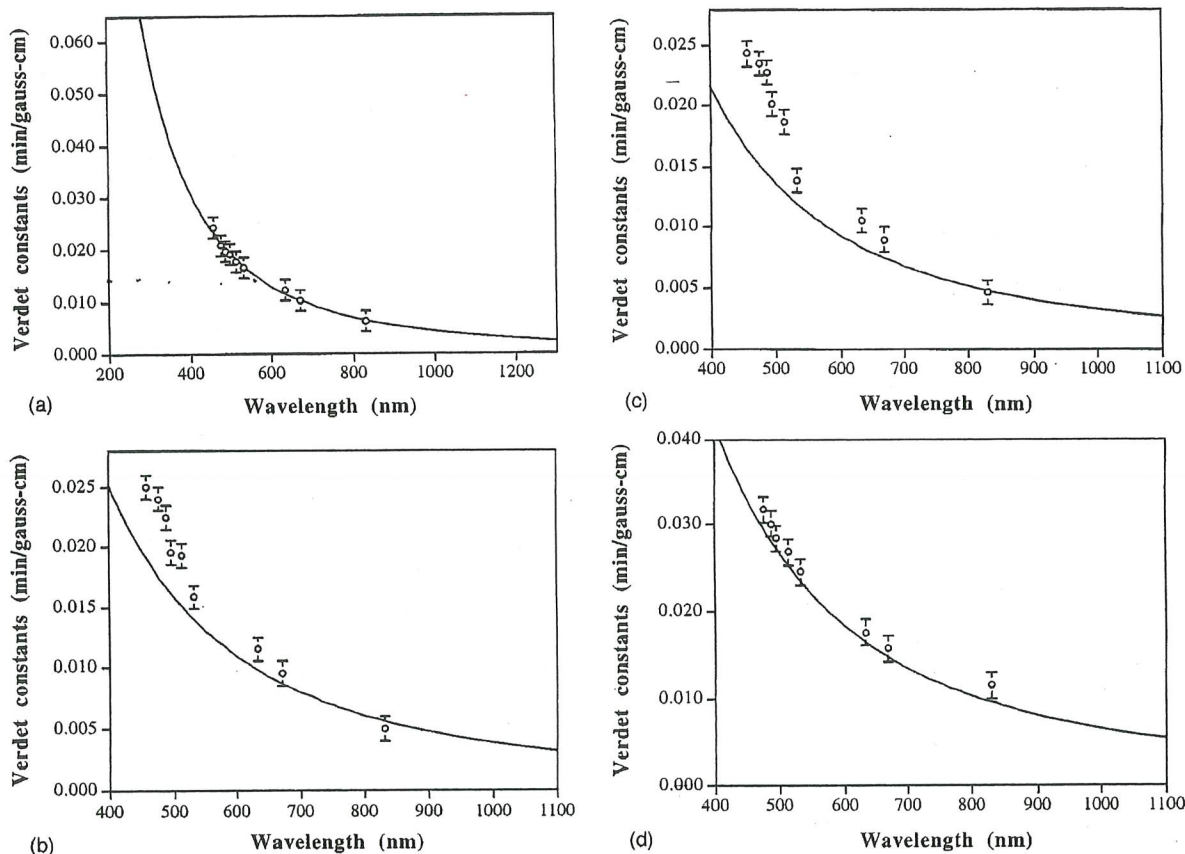


Fig. 2. The Verdet constants measured at various wavelengths along with previously published data of (a) distilled water, (b) ethyl alcohol, (c) methyl alcohol, and (d) BK7 glass, respectively. The solid lines present the power fit to previously published data and the open circles are experimental results.

stants determined from the experiment was estimated to be less than 10^{-3} min/G cm and is mainly due to the leakage of the ac magnetic field from the coils and its pickup by the photodetector. This effect is minimized by enclosing the coils in a box made of iron or moving the detector further away from the region of the ac magnetic field. It is clearly seen from Fig. 2(a) and (d) that the experimental results of distilled water and BK7 glass are in good agreement with previously published data. However, for ethyl and methyl alcohols, deviations between the experimental data [Fig. 2(b) and (c)] and the published results are observed at short wavelengths. The data are consistent with the λ^{-2} dependence⁹ of the Verdet constant for diamagnetic materials in the nonabsorbing wavelength regime.

Investigation of the concentration dependence of the Verdet constant has been carried out for aqueous ferric chloride solutions at 670 nm. The behavior of the Verdet constant is shown in Fig. 3. Our experimental data on water in Fig. 1 yield a positive value for the Verdet constant at 670 nm. In Fig. 3 we observe that the Faraday rotation changes sign from positive to negative values with increasing concentration of ferric ions. The ferric chloride solution at any particular concentration has a positive contribution to the Verdet constant from the water molecules. The ferric ions, however, are paramagnetic and make a negative contribution to the Verdet constant.⁹ Consequently, as the concentration of ferric ion increases, the Faraday rotation decreases, and at large enough concentrations becomes negative. It is interesting to

note that solutions of paramagnetic salts of appropriate concentration can provide a medium with zero or exceedingly small Faraday rotations.

V. CONCLUSIONS

An experiment suitable for a senior and graduate laboratory to measure the Verdet constant using inexpensive and

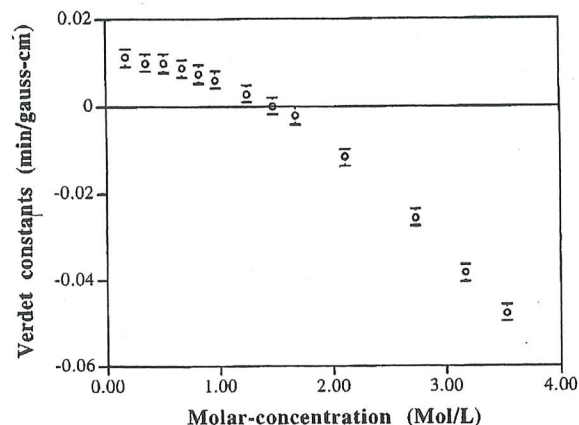


Fig. 3. Molar-concentration dependence of the Verdet constants for ferric chloride water solution measured at 670 nm.

readily available components has been developed. The Verdet constants of several commonly available materials as functions of wavelength were measured. Other than measuring Faraday rotation, this simple experiment can also be easily adapted for measuring other magneto-optic effects such as magnetic circular dichroism and the magneto-optic Kerr effect, and magnetic-field-induced changes in the gain of a cw diode laser.

ACKNOWLEDGMENTS

Helpful discussions with Dr. Xinli Jiang and Ke Yang are gratefully acknowledged.

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PHYSICS AND POETRY

Why did it take Bohm so long to work out all the implications of his theory? Here a world of explanation is necessary, for although it may seem unusual to the nonscientist, the creator of a new theory may not necessarily fully understand what he or she has done.

A scientific theory may be compared to a deeply satisfying poem. Such a poem can be visited again and again during one's life. At each reading its density, allusions, images, and resonances are unfolded. The best poetry is an inexhaustible inscape, and its metaphors suggest a variety of different readings, none of which brings closure to its range of potential meanings. In similar fashion a scientific theory exists partly as mathematical formulae and partly as a cloud of words surrounding the formulae and their variables. Like the poem, the theory's meaning, or the way it functions, can be unfolded in a variety of different ways.

F. David Peat, *Infinite Potential—The Life and Times of David Bohm* (Addison-Wesley, Reading, Massachusetts, 1997), p. 14.

A Faraday rotation experiment for the undergraduate physics laboratory

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I. INTRODUCTION

Traditional college courses in general physics cover the subject in two semesters, with the second devoted to electromagnetics, optics, and "modern" physics. It has always seemed difficult to me to convince the student that the transition from the study of electricity and magnetism to optics is a logical and smooth one as far as physical principles are concerned, despite the use of analogies and a lot of hand waving. Perhaps the performance by the student in the laboratory of the experiment discussed below will not completely achieve this goal either, but it at least does bring together fundamental elements of both electromagnetism and optics in the examination of a fundamental physical phenomenon—the rotation of plane polarized light by magnetic fields in transparent liquids.

This was first observed by Michael Faraday¹ in 1845 and is usually called the Faraday effect. The rotation is caused by circular birefringence and the relationship of the external magnetic field to this birefringence can be discussed in terms of the Zeeman effect. The phenomenon has been discussed in classical terms by several authors,² and in recent years several more modern treatments have been given.³⁻⁵ It has been used some, but not widely as the subject for an experiment in the undergraduate laboratory.^{6,7}

A schematic view of the experimental setup is shown in Fig. 1. Light from a lamp is collimated with a simple lens and aperture and directed through a color filter to make it approximately monochromatic. Next it passes through a polarizer and thus enters the liquid medium as plane polarized light. When there is current in the magnetic solenoid, a longitudinal field B exists in the liquid and the plane of polarization is rotated by an angle θ_R . This angle is given by

$$\theta_R = V \int_0^l B \cdot dl = V \overline{B}l,$$

where V is the Verdet rotation constant for the liquid being used and l is the path length traveled by the light in the liquid. V is a function of the wavelength and changes appreciable over the visible range (see Fig. 2). It is therefore appropriate to use the color filter.

After passing through the solenoid and having its polarization direction changed by θ_R , the beam traverses a second polarizer (analyzer) and is then focused onto a photodiode which acts as an intensity monitor. By adjusting the

analyzer angle for maximum intensity both with the magnetic field on and off, the rotation angle θ_R can be determined. Thus either V or $\overline{B}l$ can be measured if one or the other is already known. Using the modest apparatus described below and convenient liquids such as water or xylene, rotations of several degrees can be obtained.

II. APPARATUS

The apparatus described schematically in Fig. 1 is mounted on a simple optical bench which is 42 in. long. Let me describe the components shown in Fig. 1 by progressing from left to right.

A standard projection lamp (BEH-150 W) is used as the source. A 25- Ω , 50-W resistor is connected in series with the lamp to prolong its lifetime.

In order to provide a collimated beam of light a simple converging lens ($f = 10$ cm; diam = 51 mm) is used in conjunction with an aperture plate (3/8 in. diam. hole in a round aluminum plate). This provides a beam that is somewhat smaller in diameter than the liquid medium cell.

The color filter is a square green glass filter (5 $\times$ 5 cm; 2 mm thick) with a transmission peak at 5400 Å. One example of a suitable filter is number A26-3350 distributed by the Ealing Corporation.

The polarizer and analyzer are identical and consist of polaroid sheet sandwiched in round pieces of glass (Edmund Scientific, 40209) and mounted so that they can be rotated about an axis parallel to the light beam. Relative angles are measured with a protractor attached to the mounting.

The magnetic solenoid is wound on an aluminum spool with a bore of 5/8 in., an overall length of 7 1/2 in. and an outside diameter of 5 in. It consists of 1000 turns of #10 magnet wire and requires about 12 V to produce a current of 15 A, which is the normal operating condition. The average magnetic field along the bore is about 1100 G at 15 A, and $\overline{B}l$ is typically 22 000 G cm (0.022 T m).

The power supply has an output voltmeter and ammeter on the front panel. Current values can be maintained within a range of $\pm 1\%$ or better. We do not use the absolute value of the current for any numerical calculations of experimental results, but we do need to maintain the magnetic field at a reasonably constant level.

The optical rotation occurs in the cell containing the

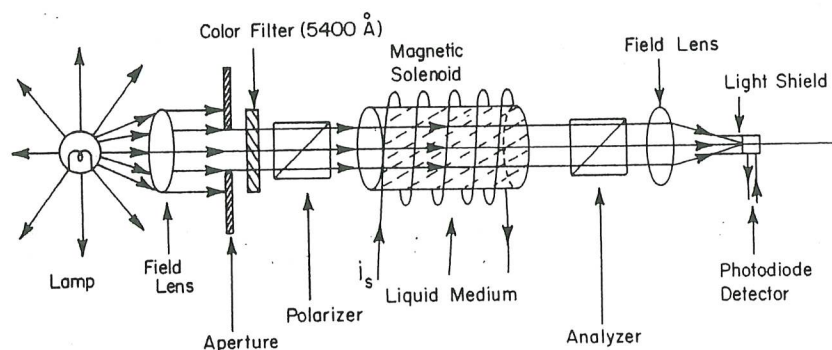


Fig. 1. Schematic diagram of the Faraday rotation apparatus.

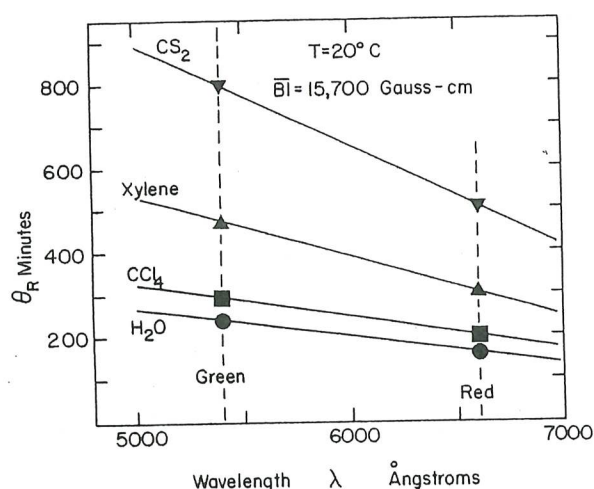


Fig. 2. Some values of the Faraday rotation angle, θ_R as measured by the author for four liquids.

active liquid medium. It consists of a bakelite tube 8 in. long, with an inner diameter of $\frac{1}{2}$ in. and outer diameter of $\frac{5}{8}$ in. The ends of the tube are sealed with $\frac{1}{2}$ in. diam. glass plugs cut from $\frac{1}{4}$ in. thick optical glass. These are glued into the tube with an epoxy resin. The tube is filled with liquid through a $1\frac{1}{2}$ in. long, $\frac{1}{4}$ in. diam. aluminum filler-expansion port which is glued into a hole drilled through the bakelite at one end of the tube.

The choice of active optical liquids was dictated by two considerations—fairly large Verdet constants and relatively nonhazardous materials. Figure 2 shows eight measurements of rotation that I made with four different liquids—water, CCl_4 , *o*-xylene, and CS_2 . These were done at two different wavelengths of light, 5400 Å and 6600 Å. The straight lines drawn on the graph are merely to guide the eye and are not intended to predict values of rotation at other wavelengths.

We use water and xylene in the laboratory. Xylene is sufficiently different from water in its optical properties that it is easy to distinguish one from the other in laboratory measurements.

The analyzer is identical to the polarizer. It of course is used to determine the amount of rotation produced by the magnetic field.

To monitor the intensity of the light transmitted by the apparatus, the beam leaving the analyzer is focussed by the last field lens ($f = 10$ cm; diam. = 51 mm) on to a photodiode. The photodiode is a VACTEC VT701. This detector, is connected in series with a 9-V battery, a 2-k Ω fixed resistor, a 10-k Ω potentiometer, and a 0–1 mA panel meter. The potentiometer allows the experimenter to set a convenient reading level on the panel meter.

III. EXPERIMENTAL TECHNIQUE

Alignment of the optical elements in the apparatus is fairly easy and is done by laboratory teaching assistants before each laboratory period. The students are encouraged to understand the alignment by making minor changes in the positions of the various elements and observing the effects.

The object of the experiment is quite simple. First the rotation of the polarization plane produced by a water sample is measured. Knowing the Verdet constant V for water one is able to determine the quantity $\bar{B}l$ for the apparatus. Next an unknown liquid sample (xylene) is measured. The

Table I. Typical rotation data for several symmetric intensity values (θ_1 and θ_2) which give measures of the angle for maximum intensity θ_m . One set of data corresponds to zero magnetic field ($i_s = 0$ A) and the other to full magnetic field ($i_s = 15$ A). In this example the angle of magnetic rotation, θ_R is found to be 5.40 deg. The optically active liquid is CCl_4 .

I/I_0	$i_s = 0$ A			$i_s = 15$ A		
	θ_1	θ_2	θ_m	θ'_1	θ'_2	θ'_m
0.3	171.5	7.5	89.5	157	10	83.5
0.4	159.5	19.5	89.5	152.5	15.5	84
0.5	149.5	28	88.75	147	20.5	83.75
0.6	140.5	38	89.25	141	27	84
0.7	129.5	49	89.25	134	34	84
			$\bar{\theta}_m = 89.25$			$\bar{\theta}'_m = 83.85$
			$\theta_R = \bar{\theta}_m - \bar{\theta}'_m = 5.40$ deg			

students are asked to identify the unknown liquid by using a table of values for V .

The technique we use for determining θ_R is somewhat detailed. Merely rotating the analyzer until a maximum intensity is observed with the detector is not a very sensitive method since the observed intensity I is related to the maximum intensity I_0 and the angle of the analyzer θ by the equation

$$I = I_0 \cos^2 \theta.$$

Thus when θ is near the angle θ_m where maximum intensity occurs one is working close to the nearly flat top of the $\cos^2 \theta$ function.

Therefore we use the following method. First the analyzer is rotated until the observed intensity is a maximum and the potentiometer in the detector is adjusted until the meter reads full scale. Then we measure the angle θ_1 and θ_2 on the analyzer protractor which give rise to equal intensity readings corresponding to symmetric points on the sides of the $\cos^2 \theta$ curve. We do this for five points corresponding to $I = (0.3, 0.4, 0.5, 0.6, 0.7)I_0$. The mean value of each set of θ_1, θ_2 measurements gives a measure of θ_m . Table I shows a typical set of data taken with the magnetic solenoid off and then energized (corresponding to magnetic solenoid current i_s values of 0 and 15 A). These data were taken by a student and it is clear from the dispersion of the measurements that one can typically determine the value of θ_R to better than 10%.

IV. SUMMARY

I am convinced that this experimental apparatus and the associated measuring techniques are well suited to the undergraduate laboratory in general physics. The kinds of measurements suggested above lend themselves to straightforward error analyses which are a very important part of undergraduate laboratory training.

One criticism that can be made of this laboratory experiment is that it is not feasible to give a detailed theoretical treatment of the Faraday rotation phenomenon within the scope of most courses in general physics. I don't feel that this is a serious shortcoming. Acquainting the student with the phenomena of nature is an important part of our job even though we are not prepared in all cases to give thorough explanations of these phenomena. For example, we present the rough ideas of the quark-lepton model of matter but hardly develop its theoretical foundations in beginning physics courses.

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ACKNOWLEDGMENTS

I want to thank E. C. Beard, E. F. Durr, and J. Meyer for help in constructing the equipment.

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FARADAY ROTATION

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This instrument is warranted for a period of **two (2) years** from the date it is delivered. For breakdowns due to defects in components, workmanship, or ordinary use, TeachSpin will pay for all labor and parts to repair the instrument to original working specifications.

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- 2) Repairs not authorized by TeachSpin, Inc. have been attempted.
- 3) The instrument has been subjected to high voltages, plugged into excess AC voltages, or otherwise electrically abused.

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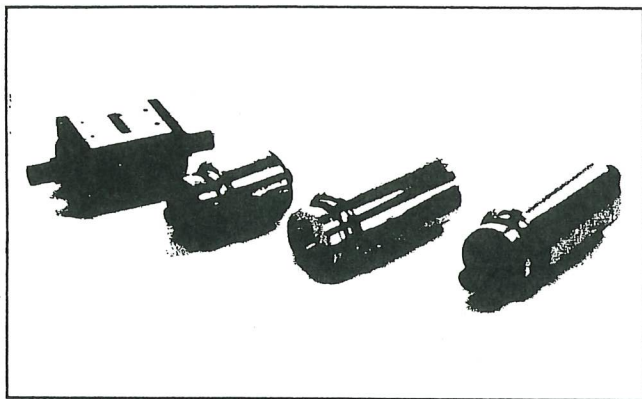
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- 2) Damage caused by operating the unit without regard to explicit instructions and warnings in the TeachSpin manual.
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FARADAY ROTATION

Commercial Applications

700 SERIES FARADAY ROTATORS AND ACCESSORIES



Introduction

The unique feature of a Faraday rotator is its nonreciprocity, that is, the fact that the "handedness" of the polarization rotation it imparts on a transmitted beam is dependent on the beam's propagation direction. This capability allows the conception of many clever schemes to manipulate laser light in ways that are otherwise impossible or, at least, much more difficult to implement. By far the most common application of Faraday rotators is their use in optical isolators, devices used to protect laser sources from harmful back reflections.

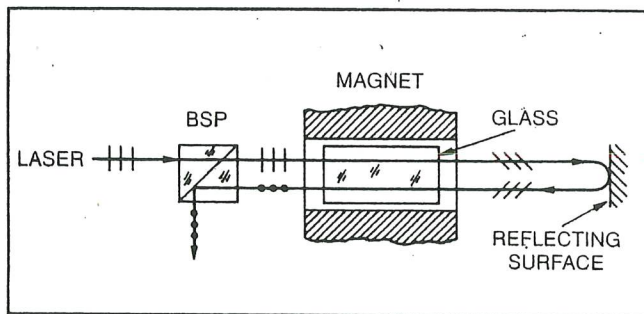


FIGURE 1: A basic optical isolator

Conoptics manufactures a family of modular Faraday rotators and interchangeable accessories which can be combined to perform various isolation functions. A basic optical isolator is shown in Figure 1. Light from a presumed well polarized laser source enters an aligned beamsplitting polarizer from the left. It then proceeds through a Faraday rotator which is tuned to 45° for the particular laser wavelength. Only one magnet is shown but, whether a single magnet or a three magnet array

is used, its major field component is parallel to the optical propagation direction. Light returning from any downstream reflecting surface undergoes an additional 45° rotation in its backward pass through the glass. It is now polarized orthogonal to the forward beam and is rejected by the polarizer.

Faraday Rotators

The Faraday rotators included in the 700 Series consist of one or more high flux NIB permanent magnets and a terbium glass element. Tunability is accomplished by moving the glass between regions of high and low field strength. An important feature of the Conoptics design is that the glass does not rotate as it is moved thereby eliminating the possibility of transmitted beam nutation due to residual wedge.

Polarization rotation is a function not only of the axial magnetic field strength, but also of the glass length, L , and its Verdet constant, V . More precisely, on axis

$$1) \Theta_F = V \int_{z_0}^{z_0+L} H_z dz$$

where z is along the propagation direct.

The value of V is strongly wavelength dependent and diminishes rapidly as wavelength increases. In order to avoid the need for extending the length of glass elements to achieve 45° rotation at long wavelengths, the local field strength must be increased by using combinations of magnets with opposing fields. The Model 713, 714, and 715, while outwardly quite large, have only a fraction of their lengths filled with glass. The magnet array design is intended to increase axial field strength but also affects the field radial uniformity. Only relatively small volumes along the axis of the magnet array offer sufficiently low gradients. Furthermore, tuning the rotator (changing z_0 in equation 1) by definition means moving the glass into a region of lower integrated axial field strength and higher gradients. It is generally recommended, therefore, that for best isolation ratio, a model be chosen in which the intended operating wavelength is near the upper end of the tuning range.

A strong absorption peak at approximately 490 nm is typical of terbium glass. At wavelengths larger than 500 nm absorption is low and the models listed are capable of operation with more than 6 watts average power. Operation at 488 nm is not

recommended but a special unit operable at 458 nm is available. TABLE 1 lists the specifications of standard Conoptics Faraday rotators. The isolation ratio (extinction) is dependent on the type of polarizer(s) used, the operating wavelength in relation to the tuning range, and beam diameter.

Polarizers

Figure 1 shows that it is actually the polarizer which performs the beam separation function in an optical isolator. It is an obvious fact that, while the homogeneity of the Faraday rotator is of critical importance and generally limits performance in single stage devices, the extinction of the isolator can be no better than the extinction of the polarizer. Conoptics offers the Model 720 polarizer in two forms; one containing a low cost, low power polarizer, the other a high performance, high power Glan prism. Extinction as high as -40dB (10^{-4}) is possible using the 720 Glan with a small beam and a properly tuned Faraday rotator. Extinction with the low power type ranges between -20 and -30dB. Both types are fully rotatable.

The polarization of light reflected back towards the laser source may not be the same as that in the forward direction. Any orthogonal polarization component accidentally produced will not be rejected by the polarizer-rotator combination shown in Figure 1. A second polarizer must be added to the assembly on the opposite end of the rotator and aligned at 45° to the original polarizer. This allows the forward beam to proceed with only the additional static loss of the second polarizer (<4%) but eliminates the unwanted polarization component in the backward direction. The modular design of 700 Series components provides for convenient mounting of polarizers to either end of rotator assemblies.

Quartz Compensators

A frequently annoying feature of optical isolators is that the polarization of the forward transmitted beam is at 45° to that of the source. Addition of a Model 730 Compensator provides a net rotation of either 0° or 90° , selectable by interchanging output and input ends of the assembly. The 730's utilize the rotatory power of Z cut crystal quartz as fixed 45° elements. The effect on a transmitted polarization form is the same as that of a Faraday rotator except it is fully reciprocal.

An additional advantage of the 730 is that it partially achromatizes the isolator, an important benefit when dealing with broadband or tunable lasers. A good approximation of the wavelength dependence of a Faraday rotator tuned to 45° at λ_0 is

$$2) \Theta_F = \pm 45^\circ \frac{\lambda^2}{\lambda_0^2} \left(\frac{\lambda_0^2 - F}{\lambda^2 - F} \right)^2$$

where F is a constant peculiar to the type of glass used and λ_0 . The sense of the rotation ($\pm$) is dependent on the direction of propagation. The rotation of the 730 compensator (45° at λ_0) is expressed similarly,

$$3) \Theta_C = 45^\circ \frac{\lambda^2}{\lambda_0^2} \left(\frac{\lambda_0^2 - C}{\lambda^2 - C} \right)^2$$

but the sense is independent of propagation direction. The net rotation of these elements in series is

$$4) \Theta = \Theta_C + \Theta_F = 45^\circ \frac{\lambda^2}{\lambda_0^2} \left[\left(\frac{\lambda_0^2 - C}{\lambda^2 - C} \right)^2 \pm \left(\frac{\lambda_0^2 - F}{\lambda^2 - F} \right)^2 \right]$$

Figure 2 is an 800-1000 nm plot of equation 4 using a Model 714 Faraday rotator and Model 730 Compensator between two polarizers. In this case λ_0 is 880 nm. Ideal components are assumed and static losses are not shown. For the sake of comparison, an uncompensated rotator between two polarizers

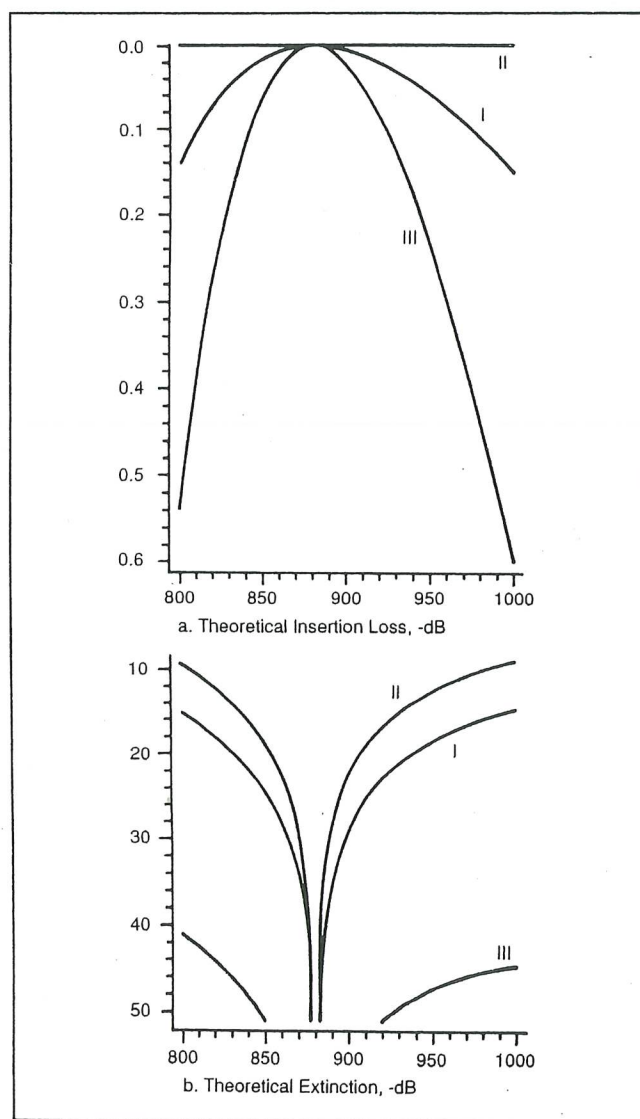
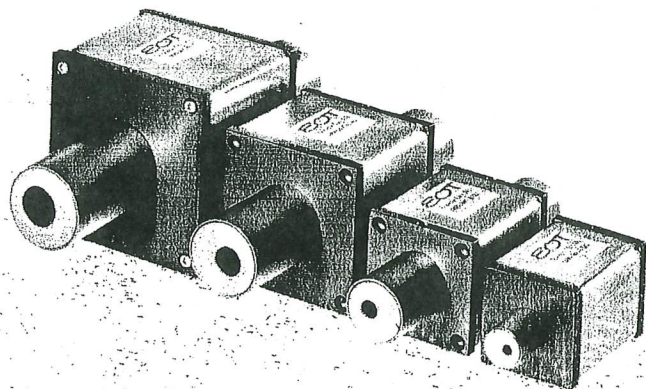


FIGURE 2: Insertion loss and extinction vs. wavelength (nm). 1, uncompensated; II, 0° rotation compensated; III 90° rotation compensated. Models 714 and 730, $\lambda_0=880\text{nm}$. Static losses are not included.

1.0 - 1.1 μ m Rotators & Isolators

Model No.	Clear Aperture	Rotation Medium	Optical Pathlength Of Rotating Medium	Rotator Transmission	Isolator Transmission	Isolation	Pulsed Damage Threshold 10nsec Pulse
1.5I1064	1.5mm	TGG & BIG	39mm	N/A	65%	≥ 60 dB	100mW cw
2XYYYY	2mm	TGG	39mm	$\geq 98\%$	$\geq 90\%$	≥ 30 dB	5J/cm ²
4XYYYY	4mm	TGG	39mm	$\geq 98\%$	$\geq 90\%$	≥ 30 dB	5J/cm ²
8XYYYY	8mm	TGG	39mm	$\geq 98\%$	$\geq 90\%$	≥ 30 dB	5J/cm ²
12XYYYY	12mm	TGG	39mm	$\geq 98\%$	$\geq 90\%$	≥ 30 dB	5J/cm ²
15XYYYY	15mm	TGG	39mm	$\geq 98\%$	$\geq 90\%$	≥ 30 dB	5J/cm ²
20XYYYY	20mm	TGG	39mm	$\geq 98\%$	$\geq 90\%$	≥ 30 dB	5J/cm ²
25XYYYY	25mm	Tb:Glass	60mm	$\geq 97\%$	$\geq 90\%$	≥ 30 dB	5J/cm ²
30XYYYY	30mm	Tb:Glass	60mm	$\geq 97\%$	$\geq 90\%$	≥ 30 dB	5J/cm ²
45XYYYY	45mm	Tb:Glass	60mm	$\geq 97\%$	$\geq 90\%$	≥ 30 dB	5J/cm ²



Ordering Information:

Model No. Identification:

- The letter X in the model numbers above is used to denote whether the device is a rotator or isolator. If a rotator is desired, use the letter R, if an isolator is desired use the letter I. Isolators come with polarizers, rotators do not.
- The letters YYYY are used to denote the wavelength. Insert your desired wavelength.

Example: An 8mm aperture isolator at 1064nm would have the model no. 8I1064.

Polarization Orientation:

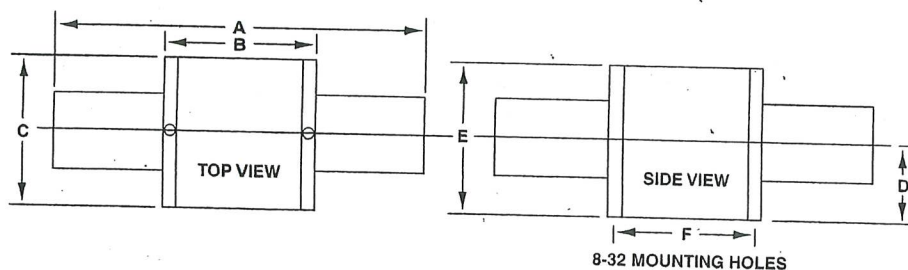
When ordering an isolator, please indicate desired input or output polarization. Our standard isolators do not include a $\lambda/2$ plate, so polarization is rotated 45°. Models with $\lambda/2$ plates are available. Contact EOT for prices.

Escape Ports:

Standard isolators do not include escape ports. Escape ports are available at no cost, but they must be specified in the order. Single and double escape ports for both the input and output polarizers are available.

Notes:

- Larger apertures are available. Call EOT for details.
- All isolators listed above use Brewster angle thin film polarizers. A small amount of beam walkoff occurs with these polarizers.



Dimensions (mm)						
Model	A	B	C	D	E	F
2XYYYY	85.2	50.3	32.1	14.6	31.8	45.6
4XYYYY	95.0	51.3	38.1	19.1	38.1	46.5
8XYYYY	126.0	51.3	50.8	25.4	50.8	46.5
12XYYYY	150.4	58.2	63.5	31.8	63.5	53.3
20XYYYY	Call EOT					

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